

EDUCATION

M.Sc. Intelligent Adaptive Systems, University of Hamburg

Oct 2021 - Feb 2025

- Fully funded by the DAAD Scholarship
- Selective (4% admission rate), highly international, research-oriented program
- *Coursework*: Computer Vision 1 & 2, Statistical Signal Processing, Intelligent Robotics, Neural Networks, Software Architecture
- *Thesis*: **Length Generalization on Multi-Digit Integer Addition with Transformers**, [PDF](#)
- *Final grade*: 1,43 (excellent), *Thesis grade*: 1,0

B.Sc. Computer Science, University of Strasbourg

Sep 2016 - Aug 2020

- Funded by the Azerbaijani Government Scholarship
- *Thesis*: Greenhouse Segmentation on Multispectral Satellite Imagery using Convolutional Neural Networks
- *Final grade*: 16.3 / 20 (mention très bien), *Thesis grade*: 19,78 / 20

PUBLICATIONS

Adaptive Recurrence as Algorithmic Time for Length Generalization in Addition

May 2026

ICML 2026 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents (CompLearn).

<https://openreview.net/forum?id=XW18V4H9sq>

SatDM: Synthesizing Realistic Satellite Image with Semantic Layout Conditioning using Diffusion Models

Sep 2023

74th International Astronautical Congress (IAC). Pre-print: <https://arxiv.org/abs/2309.16812>

Greenhouse Segmentation on High-Resolution Optical Satellite Imagery using Deep Learning Techniques

Jul 2020

Pre-print: <https://arxiv.org/abs/2007.11222>

EXPERIENCE

Data & ML Consultant

Nov 2024 — Present

Netlight Consulting

Hamburg, Germany

- Automated model training and evaluation for vessel service power demand prediction at one of the world's largest cruise lines
- Developed a document ingestion pipeline and RAG-based chatbot, including infrastructure and vector DB using AWS Bedrock

MLOps Engineer (Working Student), AI Platform Team

Feb 2023 — Aug 2024

Merantix Momentum

Berlin, Germany

- Deployed a self-hosted annotation solution for geospatial datasets, managed multi-user annotation workflow end-to-end
- Managed cloud infrastructure for a public interactive diffusion model-based installation at the World Economic Forum 2025

Research Engineer Intern, Central Research

Jul 2022 — Sep 2022

Nuance Communications (a Microsoft company)

Aachen, Germany

- Migrated speech recognition workflows (Docker containers and library code) to run on AzureML platform

Founding Engineer

Aug 2020 — Nov 2021

WeType Inc.

Baku, Azerbaijan

- Developed NLP functionality - grammar correction, spelling check, text summarization, and essay grading
- Deployed Transformer models (BERT and GPT) trained on custom datasets

Machine Learning Engineer, Research and Development

Feb 2020 — Aug 2021

Space Agency of the Republic of Azerbaijan (Azercosmos)

Baku, Azerbaijan

- Trained and deployed deep learning models that achieve 94% accuracy and 0.85 F1 score on satellite imagery segmentation

Research Intern, Research and Development

Jul 2019 — Sep 2019

Space Agency of the Republic of Azerbaijan (Azercosmos)

Baku, Azerbaijan

- Implemented SVM, random forest, and CNN models for greenhouse segmentation on satellite imagery

SKILLS

Tools and Technologies

Python, C, C++, PyTorch, Transformers, SpaCy, AWS, GCP, Docker, Terraform, Kubernetes, TensorFlow

Communication

Azerbaijani (native), Russian (native), English (fluent), German (conversational), French (rusty, but can understand)